2 Two positively charged particles X and Y of masses $3m$ and $2m$ respectively, move directly towards each other with the same speed as illustrated in Fig. 2.1.



Fig. 2.1

Fig. 2.2 shows how the velocity of each particle varies with time during the collision.

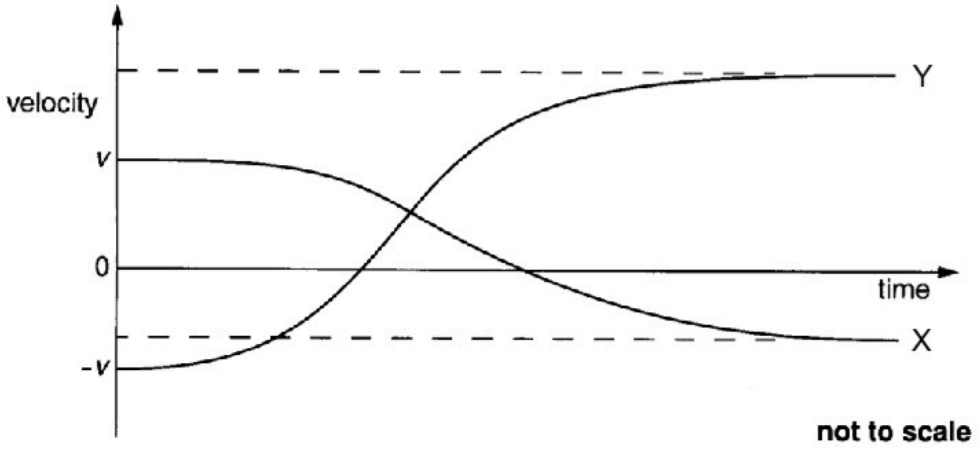


Fig. 2.2

The collision is elastic.

(a) Describe the energy change of the system that takes place during the collision.

.....

.....

.....

.....[1]

(b) Explain, using features of the graph, how it may be deduced that X has the larger mass.

.....

.....

.....

.....[2]

(c) Determine, in terms of v, the speeds of X and Y after the collision.
Show your workings.

speed of X = ...

speed of Y = ...

[4]

[Total: 7]